

CS7.303: Digital Signal Analysis

End Exam, Spring 2026

Time: **90 minutes**

Total Marks: **75**

- Answer all the questions, Marks for each question is provided in the brackets.
- Calculators are allowed.
- Clearly state the assumptions if any that were considered to solve.

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1. What are the characteristics of a linear phase filter and describe its significance precisely with an example or necessary derivations?

[5 Marks]

2. Can the convolution be implemented for any arbitrary impulse response ($h[n]$) of infinite length? Then, how an infinite length of $h[n]$ can be realizable in digital systems? Is there any constraint on $h[n]$ that can make the realization possible?

[5 Marks]

3. Show the DTFT characteristics of an ideal band-pass filter. Can this filter be physically realizable with digital systems? If yes, please justify the answer else suggest methods to overcome this.

[5 Marks]

4. A continuous time signal sampled with 16000Hz is given to an ideal digital filter (whose impulse response is $h[n]$) to suppress the components of frequencies above 4000Hz. Compute the $h[n]$.

[5 Marks]

5. Draw a flowchart to compute periodicity of a discrete-time cosine signal using correlation function?

[5 Marks]

6. Compute the DTFT of $2\cos(\frac{\pi n}{3}) + 3\sin(\frac{\pi n}{6})$?

[5 Marks]

7. For the given system whose input and output relation is $y[n] = ax[n] + c$, then check whether the system is linear or not? Show the necessary computations.

[5 Marks]

8. Why is discrete-time frequency always rational and less than 1? Justify this with necessary derivations.

[5 Marks]

9. A system with the following input-output pairs has been observed during the operation of linear system. Can you draw any conclusions about the time invariance of this system? Justify your answer clearly.

[5 Marks]

$$x_1[n] = \{-1, 2, 1\} \quad y_1[n] = \{1, 2, -1, 0, 1\}$$

$$x_2[n] = \{1, -1, -1\} \quad y_2[n] = \{-1, 1, 0, 2\}$$

$$x_3[n] = \{0, 1, 1\} \quad y_3[n] = \{1, 2, 1\}$$

10. Let $x_1[n] = \cos(\frac{2\pi n}{N})$; $x_2[n] = \sin(\frac{2\pi n}{N})$; $0 \leq n \leq N-1$, then determine the N-point circular convolution and show the necessary computations?

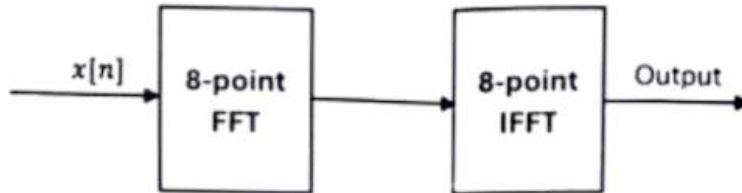
[10 Marks]

11. Let $x[n] = \{1, 2, 3, 2, 1, 0\}$ and $v[n] = \{3, 2, 1, 0, 1, 2\}$ then find the relation between DTFT of $x[n]$ and 6-point DFT of $v[n]$? Justify your answer with necessary calculations.

[5 Marks]

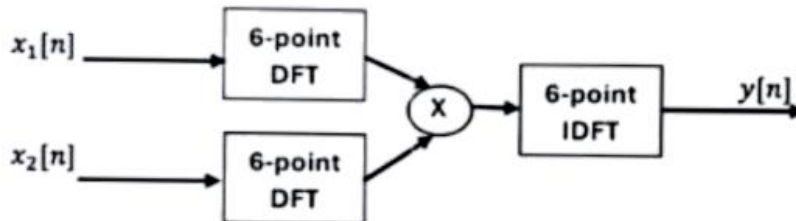
12. Let $x[n] = \{1, 2, 3, 4, 5, 6, 7, 8\}$, then compute output from the below block diagram. Justify your answer clearly.

[5 Marks]



13. Let $x_1[n]$ and $x_2[n]$ are $\{1, 2, 1\}$ and $\{1, 2, 2, 1\}$ respectively then both $x_1[n]$ and $x_2[n]$ undergone process as shown in Figure below and result $y[n]$. Find $y[n]$?

[5 Marks]



14. Let $h[n]$ and $x[n]$ are $\{1, 2, 1\}$ and $\{1, 2, 2, 1\}$, then $y[n] = x[n] * h[n]$. Write the equation involved in computing $y[2]$ and show its computation with the graphical illustration involving discrete-time signal representations?

[5 Marks]

Useful Formulas:

DTFT: $X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$

DFT: $X[k] = \sum_{n=0}^{L-1} x[n]e^{-j\frac{2\pi kn}{N}}$

IDFT: $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi kn}{N}}$
